

3.2 Pre-insertion

3.2.1 Chlorhexidine-containing body wash

- **Recommendation 1. WHO suggests using a chlorhexidine-containing body wash (liquid or impregnated wipes) daily in critically-ill adults, adolescents and children.** (Conditional recommendation, low certainty of evidence)

Remarks

- The evidence did not differentiate between liquid and impregnated wipes and evaluated chlorhexidine concentrations ranging from 2% to 4%. It was noted that most of the studies provided evidence on the use of chlorhexidine, hence a decision was made to restrict the recommendation to chlorhexidine only and to adults, adolescents and children, but not neonates (due to the risk of chemical burns) (32).
- The GDG noted that the use of this intervention would be particularly beneficial in settings with high baseline CLABSI rates. However, the GDG highlighted the need to prioritize resources at the local level based on other competing needs in the health care facility. The GDG also expressed concerns on the potential development of resistance to chlorhexidine and on waste generation when using wipes.
- Chlorhexidine products should be used with caution in patients with burns and according to the manufacturer's instructions in order to ensure they are safe for use .

Rationale for the recommendation

Overall, the GDG judged both the desirable and undesirable effects to be small, but with a balance of effects probably favouring the intervention, with a low certainty of evidence. Based on the evidence provided, the GDG opted for a conditional recommendation in favour of the intervention, in particular in the context of high CLABSI rates at baseline, where the overall benefit may be more noticeable.

Summary of the evidence

The Systematic Reviews Group identified 31 studies (10 randomized controlled trials [RCTs] (33-42) and 21 non-randomized studies of interventions [NRSIs](43-63)) comparing daily antiseptic body washing with chlorhexidine to soap and water bathing in adults, adolescents and children.

- There was low certainty of evidence that daily antiseptic body washing may result in fewer CLABSI than no routine antiseptic body washing (risk ratio [RR]: 0.67; 95% confidence interval [CI]: 0.45 to 0.99; 10 RCTs (33-42)).
- There was low certainty of evidence that routine antiseptic body washing may result in lower catheter-related mortality than no routine antiseptic body washing (RR: 0.20; 95% CI: 0.01 to 4.13; one RCT (41)).
- There was low certainty of evidence that routine antiseptic body washing may result in little to no difference in all-cause mortality when compared with no routine antiseptic body washing (odds ratio [OR]: 0.90; 95% CI: 0.76 to 1.07; five RCTs (33, 37-40)).
- There was very low certainty of evidence that routine antiseptic body washing may result in little to no difference in sepsis when compared with no routine antiseptic body washing (RR: 1.92; 95% CI: 0.66 to 5.60; one RCT (33)).

- Subgroup analyses revealed no difference in the effect for CLABSI based on age (adults versus children) or setting (ICU versus general ward).

Evidence-to-decision

A potentially reduced equity was noted due to the different availability of products around the world and their moderate costs, in particular when considering LMICs, but the intervention was considered probably cost-effective. No issues were identified in acceptability and feasibility.

Implementation considerations

- The recommendation refers to using a chlorhexidine-containing body wash (liquid or impregnated wipes) daily and not immediately prior to line insertion. The GDG acknowledges that there are variations in the available protocols on body washing and directs health and care workers to follow their national and local policies.
- The recommendation refers to all critically-ill patients that require a CVC and not limited to the ICU.
- It is advisable to engage nursing and support staff as key stakeholders in the decision (that is, time required for bathing).
- It is also advisable to engage patients and their families, considering the potential adverse effects on the skin related to exposure to chlorhexidine, including asking for any known intolerance and rare allergic reactions to chlorhexidine.
- WASH services and standards (64, 65) are essential to enable critical IPC practices, including this recommendation on body wash.

Research needs

- There is a significant variation in available protocols on how to perform patient bathing. The GDG identified this as an area of research interest to determine the most effective protocol and product to be used (that is, liquid wash versus wipes or other means, rinsing versus no rinsing, which solution to use).
- The GDG also identified the need for additional research on the use of antiseptic products other than chlorhexidine (that is, octenidine and others) using multicentre RCTs or other types of trials (that is, adaptive) and focusing on specific populations (such as haematology patients or the paediatric ICU).
- AMR and other wider implications (that is, impact on skin microbiome, cross-resistance to antibiotics and on the epidemiology of other infections caused by gram-positive and gram-negative bacteria) associated with the use of chlorhexidine and other body wash products were identified as important areas for research.
- Cost-benefit research studies including baseline surveillance data to define potential thresholds for universal use were also considered as other areas for further research.

3.3 Insertion

3.3.1 Insertion pack/kit

- **Good practice statement 6. WHO recommends that clinicians use a standardized insertion pack/kit when placing a CVC in adults, adolescents, children and neonates (Good practice statement)**

3.3.2 CVC insertion by an insertion team

- **WHO makes no recommendation regarding the use of a dedicated CVC insertion teams in adults, adolescents, children, or neonates, as the evidence remains inconclusive.**

Remarks

- A CVC insertion team is defined as consisting of individuals specifically trained and competent in CVC insertion who are dedicated to perform the insertion of CVCs.
- The search criteria used in the systematic review were inclusive as possible, including terms such as “*insertion team*”, “*vascular access team*”, “*intravenous therapy team*”.
- Establishing a dedicated CVC insertion team is an intervention that is different from ensuring specific training for all those tasked with CVC insertion, which is covered by good practice statements 1 to 3 in these guidelines.
- Consideration should also be given to specific populations (i.e., neonates, children, or patients with known difficult intravenous access) as the potential benefit in preventing CLABSI may be confined to these groups.
- The GDG members acknowledged the difficulties in designing a RCT to evaluate this topic and the very limited available evidence.

Rationale for the decision

Based on the evidence provided (two uncontrolled before-after studies and no RCTs), the GDG considered both the desirable and undesirable effects to be small. The GDG judged that the crucial factor was the adequate training of the clinician inserting the CVC, rather than having a dedicated team itself. For this reason, the GDG considered that the balance of effects did not favour either the intervention (a formal insertion team) or the comparison (insertion by a clinician who is not part of a formal insertion team), provided that the clinician has undergone appropriate training and demonstrated competency in CVC insertion. Another concern expressed by the GDG was the potential loss of knowledge and skills in CVC insertion among non-team members (doctors and nurses), such that their insertion competency could decline. This may be especially impactful in emergency situations where they are required to urgently insert a CVC without the assistance of a formal insertion team, with potential implications affecting the wider hospital capacity and patient care. Overall, given the very low certainty of the evidence, no recommendation could be made.

Summary of the evidence

- The Systematic Reviews Expert Group did not identify any RCTs on the topic, but only two uncontrolled before-after studies (66, 67) evaluating the use of a dedicated CVC insertion (for non-haemodialysis) team compared with not using a dedicated CVC insertion team. The risk of bias in both studies was rated as high due to their uncontrolled design (no formal assessment).
- There was very low certainty of evidence from one study in children and adults reporting a 58% reduction in the number of CLABSIs after the introduction of a highly specialized vascular access team (67). Similarly, there was very low certainty of evidence from another study in neonates reporting that overall CLABSI rates decreased by 65% after implementation of the line team (66).
- There were no studies identified reporting on catheter-related mortality, all-cause mortality, sepsis pneumothorax, local site infections, bleeding events that require action, cardiac tamponade, deep vein thrombosis or venous stenosis.

Evidence-to-decision

Increased moderate costs (due to the resources required to maintain a dedicated team), the probably reduced equity, variable acceptability by health care workers and patients (hospital management might not be in favour due to the additional costs, but patient satisfaction is likely to be improved) and low feasibility (as having a dedicated team is not always possible due to staff shortages) were all considered, resulting in a decision to make no recommendation.. The overall cost-effectiveness judgement did not favour either the intervention or the comparison.

Research needs

- Further research is needed to compare the potential cost-benefit of using a formal CVC insertion team versus an educational/behavioural programme for clinicians inserting CVC, including both ICU and non-ICU settings.

3.3.3 Sterile technique

- **Good practice statement 7. WHO recommends using a sterile technique for the insertion of all CVCs in adults, adolescents, children and neonates.**
(Good practice statement)
- **Good practice statement 8. WHO recommends that clinicians use sterile single-use gloves compared to non-sterile gloves when inserting a CVC in adults, adolescents, children and neonates.**
(Good practice statement)
- **Recommendation 2. WHO suggests using a sterile technique involving maximal sterile barrier precautions over a sterile technique not involving maximal sterile barrier precautions at the time of CVC insertion in adults, adolescents, children and neonates.**
(Conditional recommendation, very low certainty of evidence)

Remarks

- *Maximal sterile barrier precautions*: sterile barrier precautions that include the use of a cap, mask, sterile gown, sterile gloves and a sterile full body drape (11).
- *Sterile technique* is defined as a set of specific practices and procedures performed to make equipment and areas free from all microorganisms and to maintain that sterility (11).
- GDG members acknowledged that the above definition of sterile technique as comparator also seems to be generic and it may not provide enough practical details.

Rationale for the recommendation

There was an extensive discussion during which a number of GDG members expressed the opinion that a strong recommendation would have been more suitable; others suggested a conversion into a good practice statement. However, it was decided to assess the available evidence and to proceed with the evidence-to-decision framework. Following the assessment, the GDG judged the desirable and undesirable effects to be small and trivial, respectively, but with a balance of effects probably favouring the intervention, despite the very low certainty of evidence. In view of the data presented, the GDG members opted for a conditional recommendation in favour of maximal sterile barrier precautions, but acknowledging that the current evidence is weak and more studies are needed to assess if less barriers may be sufficient.

Summary of the evidence

The Systematic Reviews Expert Group identified eight studies (two RCTs (68, 69) and six NRSIs (70-75)) comparing maximal sterile barrier precautions in adults, children and adolescents, and neonates.

- There was very low certainty of evidence that maximal sterile barrier precautions may result in fewer CLABSI than a lesser standard of precautions (RR: 0.47, 95% CI: 0.10 to 2.22; two RCTs (68, 69)).
- Two non-randomized studies also reported very low certainty of evidence that maximal sterile barrier precautions may result in fewer CLABSI than a lesser standard of precautions (RR: 0.21; 95% CI: 0.05 to 0.94; two non-RCTs (70, 71)).
- There was very low certainty of evidence that maximal sterile barrier precautions may result in little to no difference in local insertion site infections when compared with a lesser standard of precautions (RR: 1.01; 95% CI: 0.06 to 16.03; one RCT (68)).
- There were no studies identified that reported on catheter-related mortality, all-cause mortality, sepsis and other complications.

Evidence-to-decision

The intervention may increase costs due to the additional equipment required and it is probably accepted and feasible as it is already common practice. However, patients may have more discomfort with full body drapes. In addition, the GDG judged equity to be probably increased.

Implementation considerations

- Ensure adequate training, understanding and competency in the application of maximal sterile barrier precautions.
- The GDG discussed the need for a dedicated procedure room for line insertion and noted that this is not practical for critically-ill patients and may be an issue in some health care facilities, particularly in LMICs. Furthermore, it is not part of the current definition of maximal sterile barrier precautions.

Research needs

- More research studies are needed to assess if lesser sterile precautions are needed, with a trial design and platform (for example, adaptive (76, 77)) that would allow to clearly distinguish the impact of the different components within maximal sterile barrier precautions (that is, cap, mask, full body drape, the environment where the line is placed) and allow to answer multiple questions simultaneously.

3.3.4 Skin disinfection

- **Good practice statement 9. WHO recommends that skin disinfection should always be used prior to the insertion of all CVCs in adults, adolescents, children and neonates.**
(Good practice statement)
- **Recommendation 3. WHO suggests using an alcohol-based formulation over an aqueous-based antiseptic formulation for skin disinfection before CVC insertion in adults, adolescents, children, and neonates.**
(Conditional recommendation, low certainty of evidence)

Remarks

- Alcohol-based antiseptics should be the preferred choice for skin disinfection before CVC insertion.
- It was acknowledged that alcohol-based formulations can cause skin irritation, dryness and a risk of pain in case of laceration and these potential side-effects should be taken into consideration when using these products. Some of these side-effects may be caused by either the alcohol or the antiseptic component.
- Alcohol-based antiseptics can cause skin lesions in preterm infants without intact skin.

Rationale for the recommendation

The GDG considered alcohol-based formulations as the preferred option for skin disinfection over aqueous formulations, with a conditional recommendation due to the low certainty of evidence. The GDG considered the desirable effects of alcohol-based formulations for skin disinfection to be moderate and the undesirable effects trivial, with a balance of effects probably favouring the intervention.

Summary of the evidence

The Systematic Reviews Expert Group found data from five RCTs (78-82).

- There was low certainty of evidence that alcohol-based solutions may result in fewer CLABSI than aqueous-based solutions for skin disinfection (RR: 0.59; 95% CI: 0.36 to 0.98; five RCTs (78-82)).
- There was very low certainty of evidence that alcohol-based solutions may result in less all-cause mortality than aqueous-based solutions for skin disinfection (RR: 0.79; 95% CI: 0.48 to 1.30; two RCTs (78, 79)).
- There was very low certainty of evidence that alcohol-based solutions may result in fewer sepsis episodes when compared with aqueous-based solutions for skin disinfection (RR: 0.68; 95% CI: 0.35 to 1.30; one RCT (78)).
- There were no studies identified that reported on catheter-related mortality, pneumothorax, local insertion site infections and other complications.